

KIWI 4K FPGA BOARD

Gowin FPGA 4K LUTs, Cortex-M3 Hard-core

KIWI NANO 4K FPGA USER GUIDE

14/11/2025

KIWI 4K FPGA BOARD

Gowin FPGA 4K LUTs, Cortex-M3 Hard-core

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Revision History

Date	Version	Description
14/11/2025	1.0	Initial version published

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1 About This Guide

1.1 Purpose

The Kiwi nano 4k V1.0 development board user manual consists of the following four parts:

- A brief introduction to the features and hardware resources of the development board
- An introduction to the hardware circuits functions, circuit, and pinout
- Considerations for the use of the development board
- Introduction to the use of the FPGA development software.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com

- <https://cdn.gowinsemi.com.cn/UG863E.pdf>
- <https://cdn.gowinsemi.com.cn/DS861E.pdf>
- https://www.gowinsemi.com/upload/database_doc/1807/document/684c55d26c6dd.pdf
- https://www.gowinsemi.com/upload/database_doc/1811/document/6738b941b4383.pdf
- <https://cdn.gowinsemi.com.cn/UG290E.pdf>
- <https://cdn.gowinsemi.com.cn/SUG100E.pdf>

1.3 Abbreviations and Terminology

Abbreviations and Terminology	Full Name
BSRAM	Block SRAM
DLL	Delay-locked Loop
DSP	Digital Signal Processing
FPGA	Field Programmable Gate Array
GPIO	General Purpose Input Output
LDO	Low Dropout Regulator
LED	Light Emitting Diode
LQ100	LQFP100
LUT4	Four-input Look-up Table
PLL	Phase-locked Loop
SSRAM	Shadow SRAM

Table 1 Table Of abbreviations and terminology

1.4 Support and Feedback

Onekiwi Technology provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

- Website: <https://onekiwi.com.vn/>
- Email: support@ontech.com.vn

2 Development Board Description

2.1 Overview



Figure 2-1 Front view of Development board

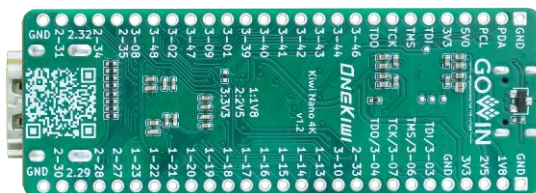


Figure 2-2 Back view of Development board

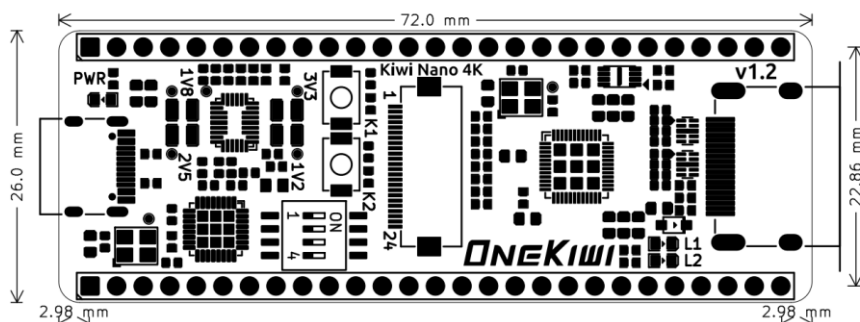


Figure 2-3 Kiwi Nano 4k Drawing

The OneKiwi Nano-4K development board is built around the **GW1NSR-4C** device in a compact QFP48 package. As part of GOWIN's GW1NSR family, this device integrates a low-power FPGA fabric together with a **built-in ARM Cortex-M3 hard processor**, embedded SRAM, and non-volatile configuration flash. This combination provides fast start-up, high reliability, and a powerful SoC-style architecture in a small and cost-effective package.

The Nano-4K board supports **multiple user-selectable I/O voltages**, including **1.8 V, 2.5 V, and 3.3 V**, enabling direct interfacing with a wide range of sensors and peripherals. The board also includes essential development features such as GPIO expansion pins, user LEDs, push-buttons, a **27 MHz system clock**, and an integrated **USB-to-JTAG/serial interface** that allows FPGA and ARM-core programming without additional hardware.

In addition to its general logic-development capability, the Nano-4K board provides a **camera interface** for entry-level image-acquisition applications and an **HDMI output** suitable for simple video-display experiments. These features enable

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beginners and advanced users alike to explore combined FPGA-and-ARM video processing, display control, and small multimedia projects.

2.2 A Development Board Suite

A development board suite includes the following items:

- Onekiwi_GW1NSR_4C_QFP48_V1.0 development board
- USB type C
- Camera OV2640
- HDMI wire and Screen

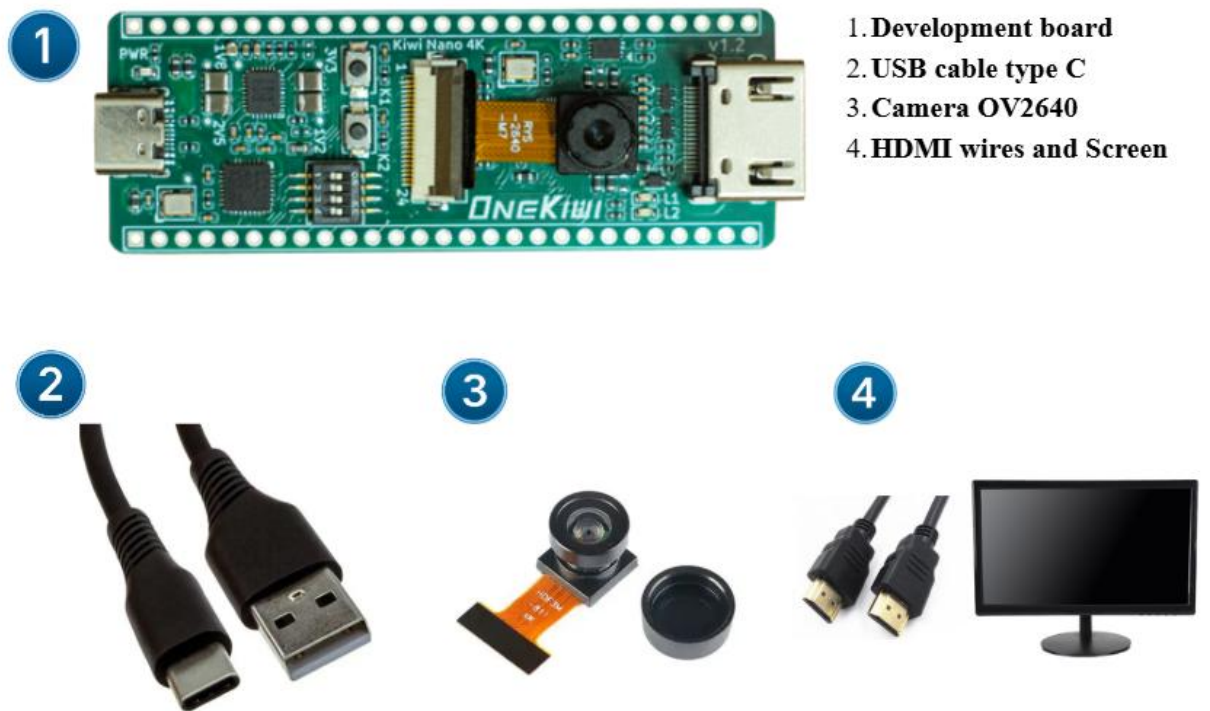
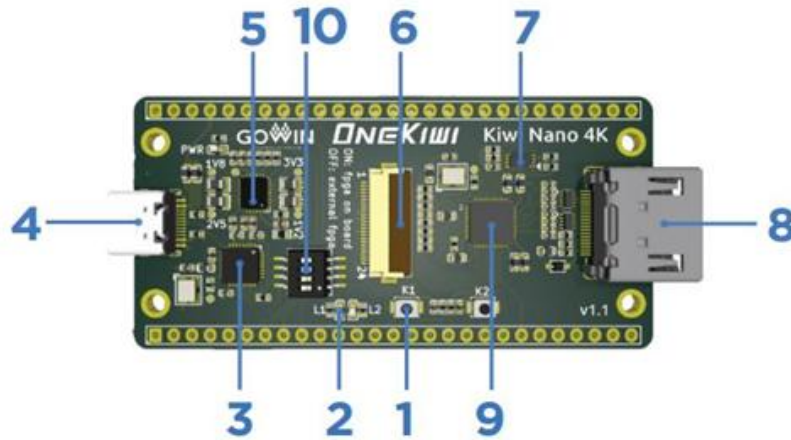


Figure 3 A development board suite

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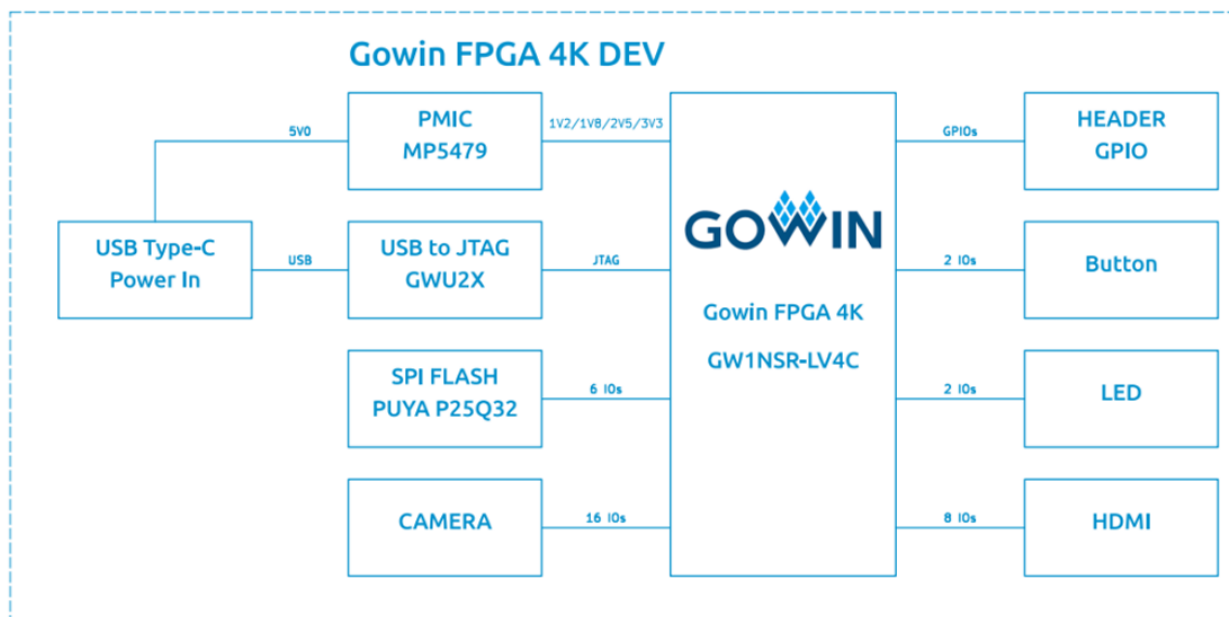
2.3 PCB components



- | | |
|-------------------|----------------------|
| 1. PUSH BUTTON | 6. Camera Connector |
| 2. LED | 7. NOR Flash |
| 3. GOWIN GWU2X IC | 8. HDMI Connector |
| 4. USB-JTAG PORT | 9. GW1NSR-LV4C |
| 5. PMIC MP5479 | 10. JTAG Port Switch |

Figure 2-4 Onekiwi nano 4k Components

2.4 System Architecture



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2.5 Features

- **FPGA CORE**
 - GW1NSR-4C QFN48PC6/I5 FPGA
 - Built-in Flash,
 - LUT4 + BSRAM
 - GPIOs
-
- **POWER SYSTEM**
 - 5V Input
 - 3.3V, 2.5V, 1.8V (I/O, interface)
-
- **CLOCK**
 - 27 MHz Oscillator → FPGA
 - 12 MHz Oscillator → USB GWU2X Interface
-
- **USB INTERFACE**
 - GWU2X chip
 - USB-C Connector
-
- **GPIO HEADERS**
 - 2 × 24-pin headers
 - 30+ GPIOs
-
- **USER INPUT + OUTPUT**
 - Camera MIPI OV2640
 - HDMI
 - 2× Keys
 - 3× LEDs
 - Power
 - 2× User
-
- **SPI FLASH**
 - 64Mbit MSPI Flash for config
-
- **JTAG**
 - TCK, TMS, TDI, TDO via header

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2.6 Development Board Specification

		Kiwi Nano 4K
FPGA Details	FPGA Chip	GW1NSR-LV4C
	Logic Units	4608
	Register	3456
	Hard processor	ARM Cortex-M3
	Block SRAM (bits)	180K
	User Flash (bits)	256K
	Number of PLL	2
	Number of I/O Bank	4
	Number of user I/O	44
User Interface	Screen Interface	HDMI
	Camera Interface	DVP Interface
	User Button	02
	User LED	02
Programmer	Programmer IC	Gowin U2X Support Programming for External FPGA (Change Switch SW1)
Physical	Size	72mm x 30 mm (Rev 2.0)
	Interface	Double row header pitch 2.54mm
Power	Main Voltage	5V
	I/O Voltage	Bank 0 & Bank 1: 3.3V Bank 2: 2.5V Bank 3: 1.8V
Environment	Temperature	Industrial: -40°C to 75°C
	Humidity	Humidity (Non-condensing): 10% - 90%

Table 2 Kiwi 4K nano specification

3 FPGA Circuits

3.1 FPGA Module

Overview

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For the resources of GW1N series of FPGA Products, refer to <https://cdn.gowinsemi.com.cn/DS861E.pdf>

I/O Bank Introduction

For the I/O BANK, package, and pinout information, see <https://cdn.gowinsemi.com.cn/UG863E.pdf> for more details.

3.2 Download Interface

3.2.1 Overview

The development board provides an USB download interface. The data stream file can be downloaded to the internal SRAM, internal flash, or external flash as needed.

Note!

- When downloaded to SRAM, the data stream file will be lost if the device is power down, and it will need to be downloaded again after power-on.
- If downloaded to flash, the data stream file will not be lost if the device is powered down.

3.2.2 USB Download Circuit

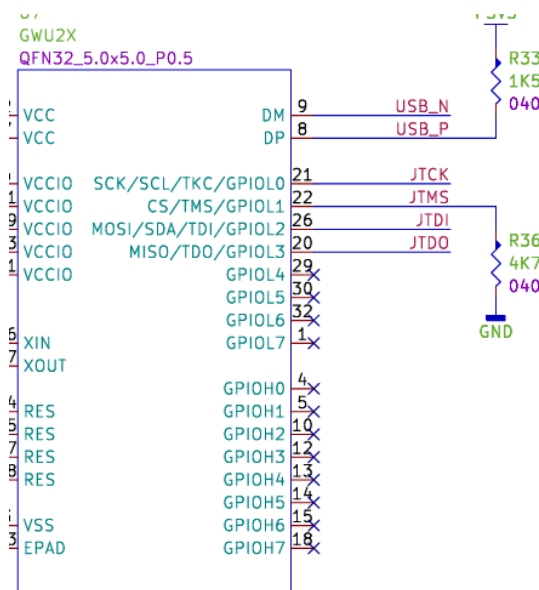


Figure 3-1 GWU2X USB TO JTAG Chip

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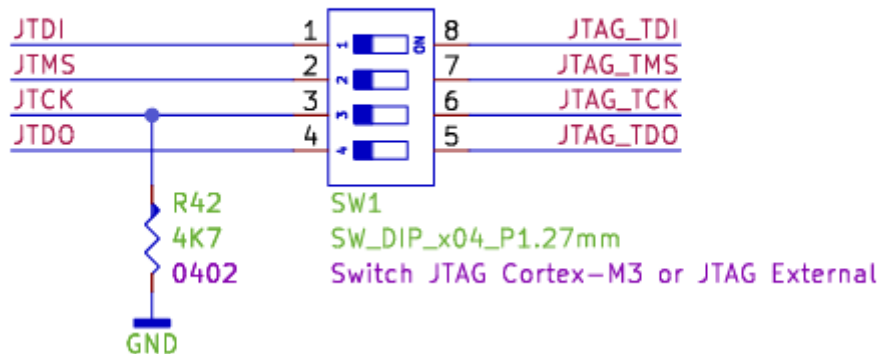


Figure 3-2 DIP Switch support extenal Dowloading

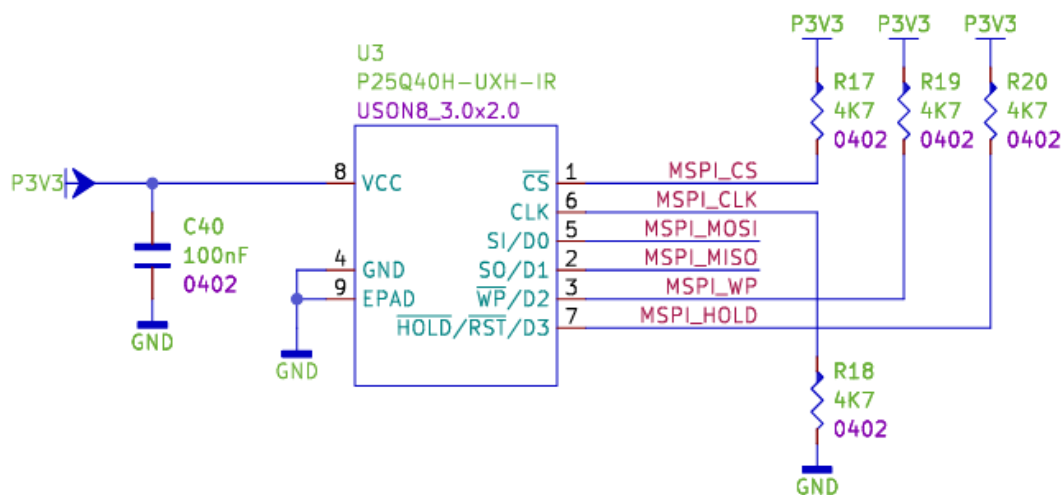


Figure 3-3 SPI flash downloading

3.2.3 Downloading the Data Stream

The data stream file can be downloaded in the following ways.

1. SRAM

Scan the device and download the bit file after powering the device on. The Done indicator lights up to denote that the download has been successful.

2. Internal Flash

Power on and download. After downloading the data stream file successfully, power down to reset and load the bit file from the internal Flash, and when the Done indicator lights up to denote that the download has been successful.

3. External Flash Configuration

Power on and download. Power down to reset and load the bit file from the external flash. The Done indicator lights up to denote that the download has been successful.

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3.2.4 Pinout

Signal Name	Pin No.	BANK	Description	I/O
JTMS	6	0	JTAG Signal	VCCIO0
JTCK	7	0	JTAG Signal	VCCIO0
JTDI	3	0	JTAG Signal	VCCIO0
JTDO	4	0	JTAG Signal	VCCIO0
MSPI_CLK	1	0	FLASH signals configuration	VCCIO0
MSPI_CS	2	0	FLASH signals configuration	VCCIO0
MSPI_WP	8	0	FLASH signals configuration	VCCIO0
MSPI_HOLD	9	0	FLASH signals configuration	VCCIO0
MSPI_MISO	47	1	FLASH signals configuration	VCCIO1
MSPI_MOSI	48	1	FLASH signals configuration	VCCIO1

Table 3 Pinout Downloading

3.3 Power Supply

3.3.1 Overview

The 5V input power interface has overcurrent and inverse current protection. The overcurrent limit is 2A.

The MP5479 power supply chip is used to step down voltage from 5V to 3.3V, 1.2V, 2.5V, 1.8V. The power supply can support up to 2A, which can meet the power demand of the development board.

3.3.2 Power System Distribution

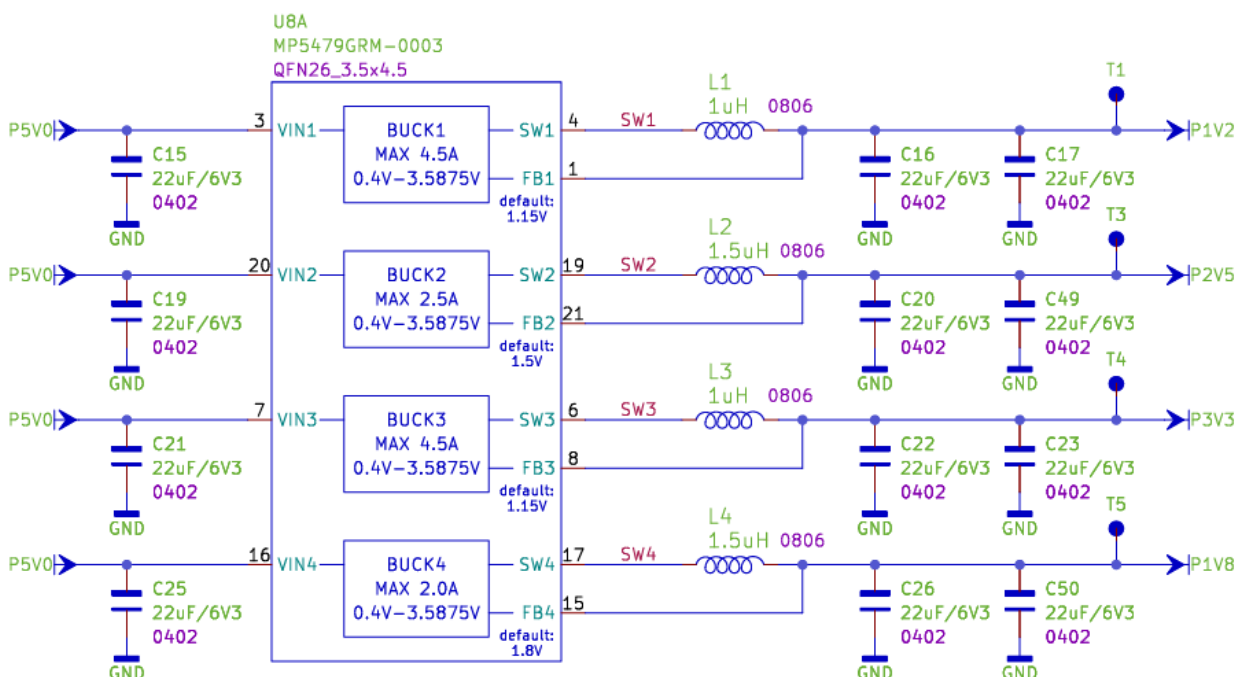


Figure 3-4 Power distribution

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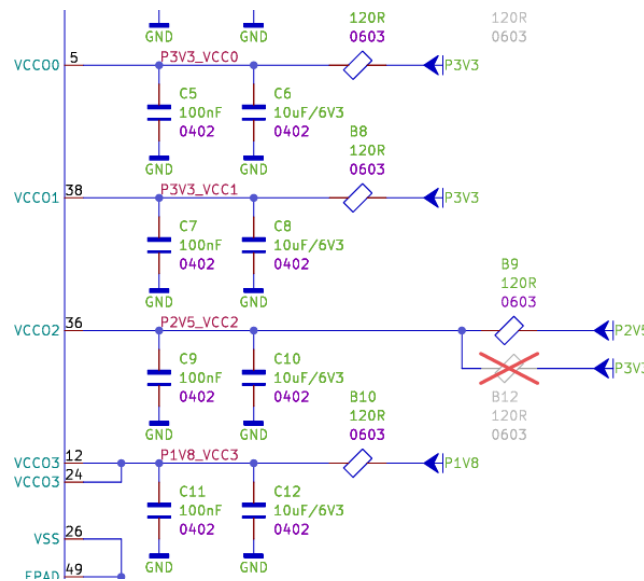


Figure 3-5 Power banks

3.4 Clock

3.4.1 Overview

A 27MHz crystal oscillator is provided in the development board that connects to the PLL input pin. This can be employed as the input clock for the PLL in FPGA, and the output clock as needed via multiplication and division of the PLL frequency.

To facilitate testing, a SMA socket is reserved on the development board as the clock input interface. The clock signal is connected to the FPGA global clock pin.

3.4.2 Clock

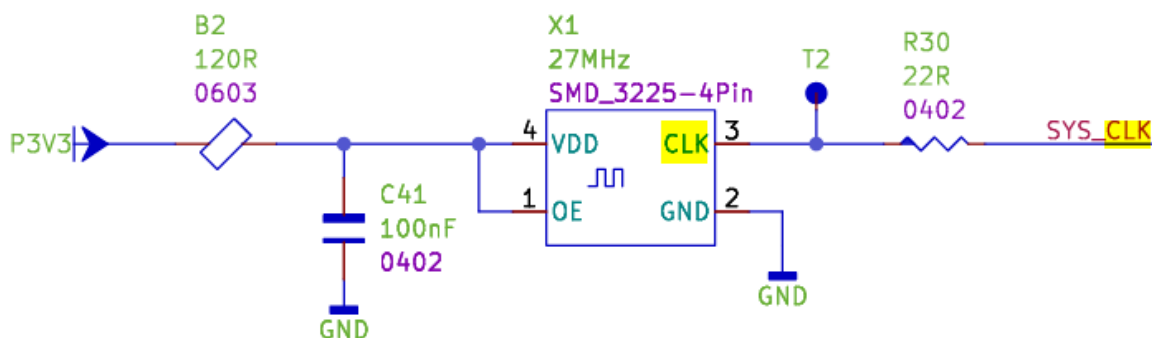


Figure 3-6 Clock schematic

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3.4.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
SYS_CLK	45	1	27 MHz crystal oscillator input	3.3V

Table 4 Pinout Clock

3.5 LED

3.5.1 Overview

Two green LEDs are incorporated into the development board and are used to display the required status. In addition, two LEDs are reserved to signify power supply and FPGA loading status.

Users can test the LEDs in the following ways:

- If the output signal of related pins is logic low, LED is on.
- If logic is high, LED is off.

3.5.2 LED Circuit

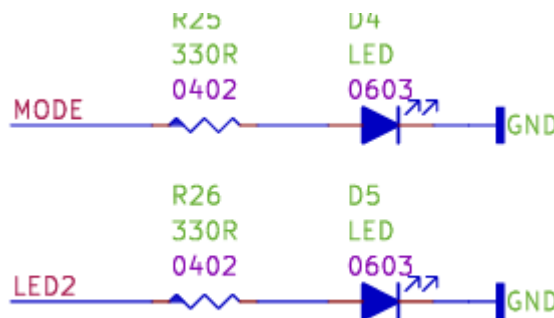


Figure 3-7 Led circuit

3.5.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
MODE	10	0	LED1	3.3V
LED2	13	3	LED2	1.8V

Table 5 Led pinout

3.6 Key

3.6.1 Overview

Two key switches are embedded in the development board. Users can manually input a low level to the corresponding FPGA pins for testing purposes.

3.6.2 Key Circuit

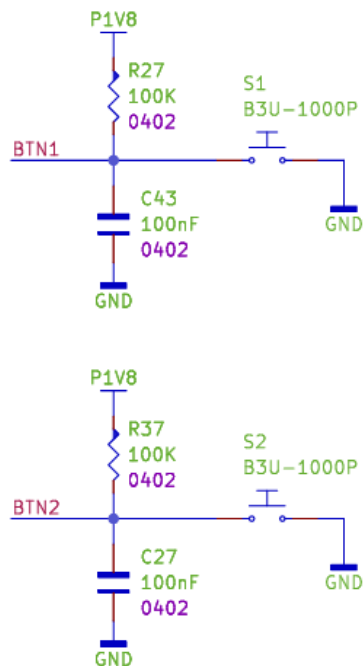


Figure 3-8 Key circuit

3.6.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
BTN1	14	3	KEY1	1.8V
BTN2	15	3	KEY2	1.8V

Table 6 Key pinout

3.7 HDMI

3.7.1 Overview

The OneKiwi Nano 4K board integrates a high-performance HDMI interface designed to support high-resolution video output for display, multimedia, and embedded imaging applications

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3.7.2 HDMI circuit

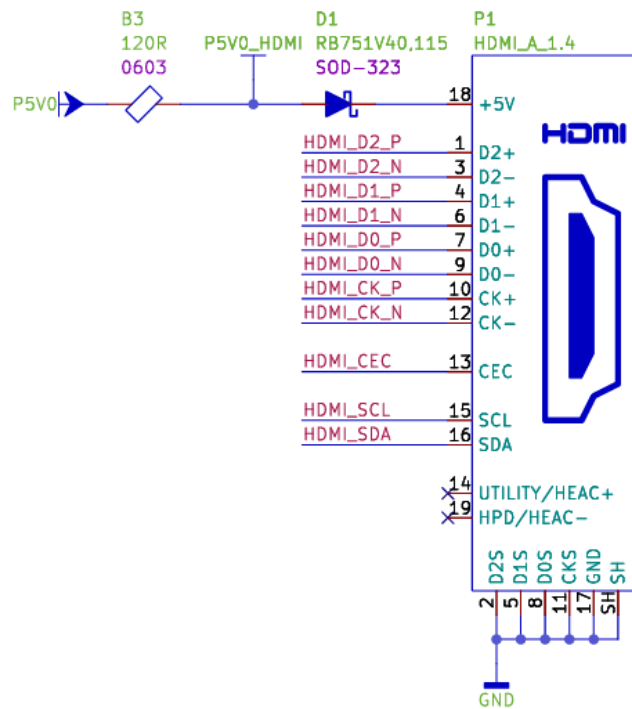


Figure 3-9 HDMI circuit

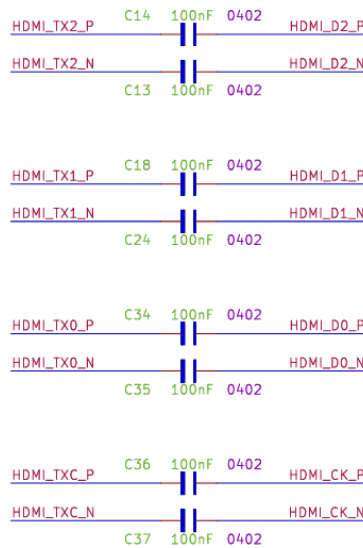


Figure 3-10 HDMI circuit 2

3.7.3 Pinout

Signal Name	Pin No.	BANK	I/O
HDMI_TX2_P	35	2	VCCIO2
HDMI_TX2_N	34	2	VCCIO2
HDMI_TX1_P	32	2	VCCIO2
HDMI_TX1_N	31	2	VCCIO2
HDMI_TX0_P	30	2	VCCIO2

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Signal Name	Pin No.	BANK	I/O
HDMI_TX0_N	31	2	VCCIO2
HDMI_TXC_P	28	2	VCCIO2
HDMI_TXC_N	27	2	VCCIO2

Table 7 HDMI Pinout

3.8 Camera

3.8.1 Overview

The OneKiwi Nano-4K development board supports the OV2640 camera module through its integrated DVP (Digital Video Port) interface. The OV2640 is a compact, low-power 2-megapixel image sensor that provides JPEG and RAW output formats, making it well-suited for embedded vision and image-processing applications. The module communicates with the FPGA using an 8-bit parallel data bus (DVP_D0–D7), along with synchronization signals including PCLK, VSYNC, and HSYNC. The I2C bus (SCL/SDA) is used for sensor configuration, allowing full control of resolution, frame rate, image effects, and exposure settings.

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3.8.2 Camera Circuit

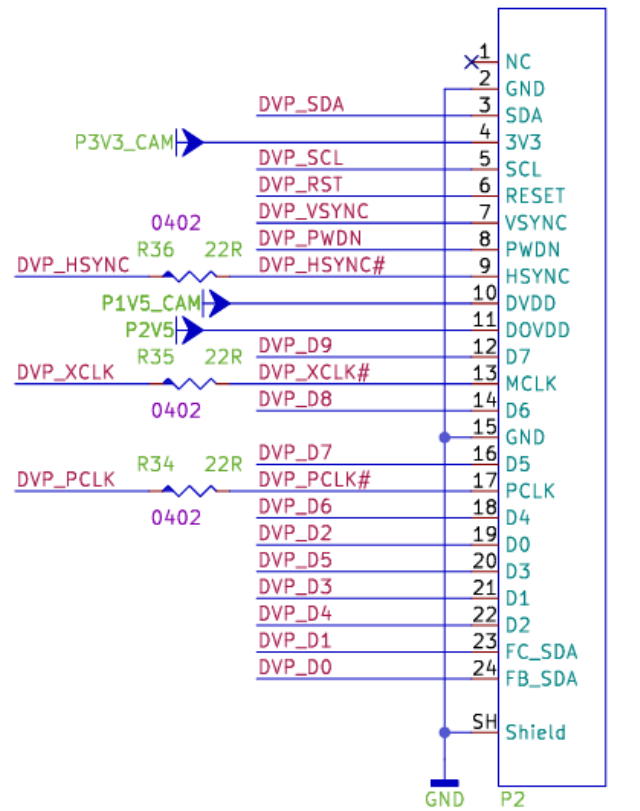


Figure 3-11 Camera circuit

3.8.3 Pinout

Signal Name	Pin No.	BANK	I/O
DVP_D0	22	3	VCCIO2
DVP_D1	21	3	VCCIO2
DVP_D2	17	3	VCCIO2
DVP_D3	19	3	VCCIO2
DVP_D4	20	3	VCCIO2
DVP_D5	18	3	VCCIO2
DVP_D6	16	3	VCCIO2
DVP_D7	23	3	VCCIO2
DVP_XCLK	33	2	VCCIO2
DVP_D8	39	1	VCCIO1
DVP_D9	40	1	VCCIO1
DVP_PCLK	41	1	VCCIO1
DVP_HSYNC	42	1	VCCIO1
DVP_VSYNC	43	1	VCCIO1
DVP_SCL	44	1	VCCIO1
DVP_SDA	46	1	VCCIO1

Table 8 DVP Camera Pinout

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3.9 GPIO

3.9.1 Overview

Two 2.54mm 24P sockets are reserved on the development board for user function extension and testing purposes.

3.9.2 GPIO Circuit

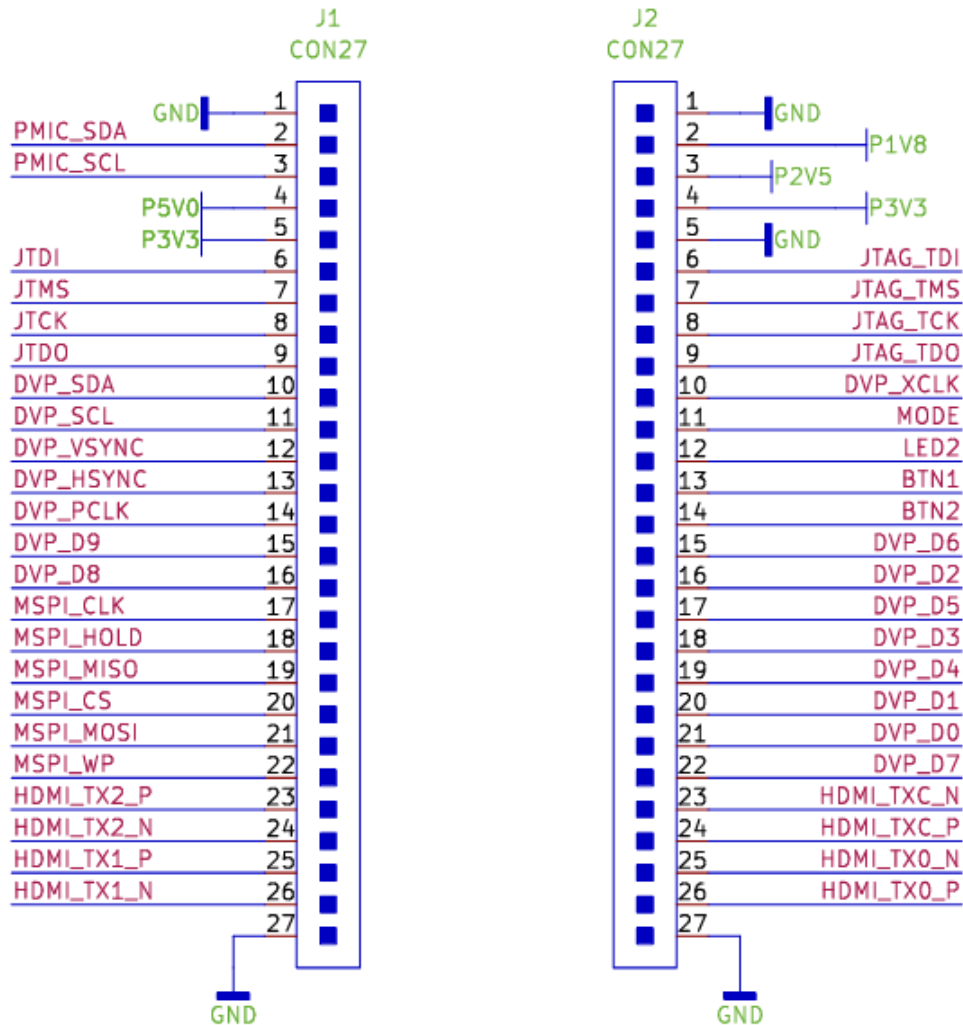
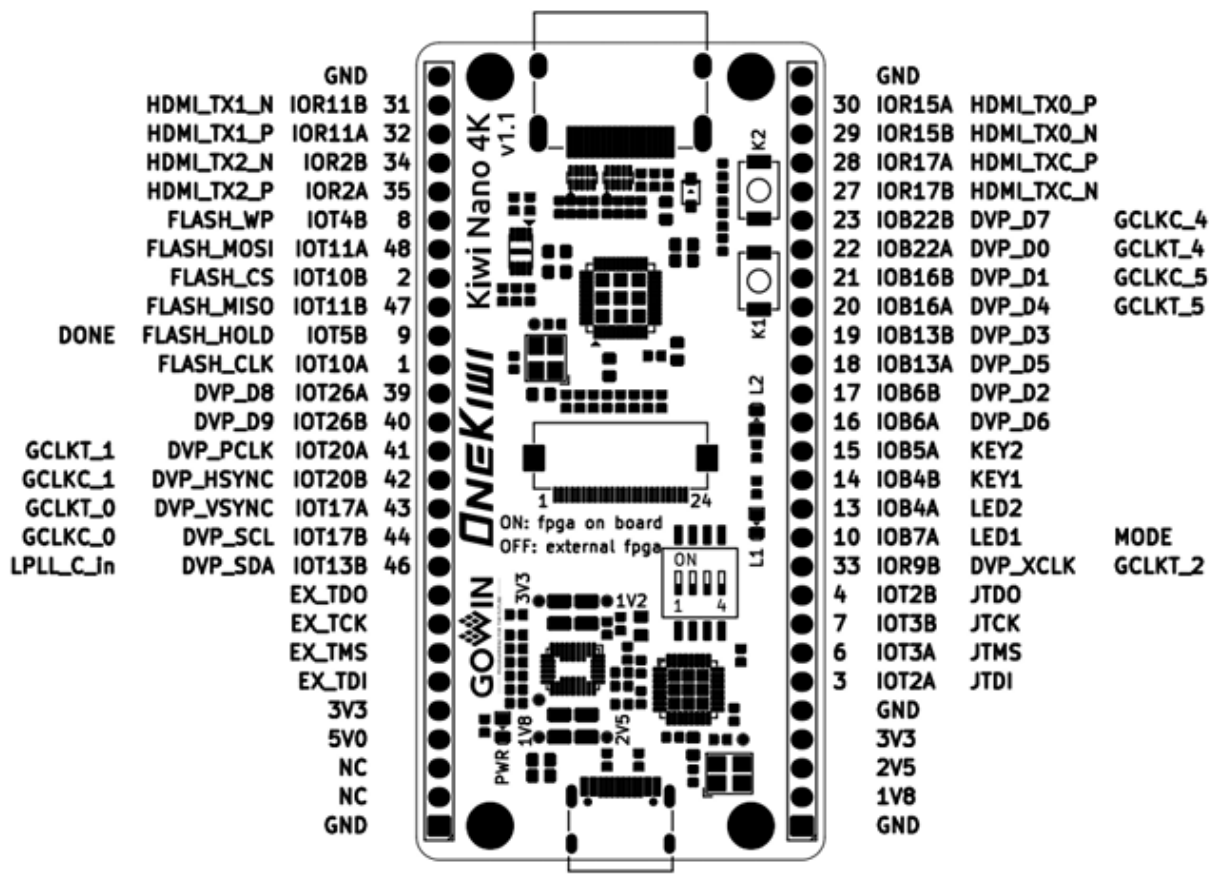


Figure 3-12 GPIO circuit

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3.9.3 Pinout



3.10 Summary Pinout for Kiwi nano 4k



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Category	Kiwi Nano4K	Signal Name	Pin No.	Pin Name	Alternate Function	Pull-up
	GND					
	NC					
	NC					
	5V0					
	3V3					
	EX_TDI					
	EX_TMS					
	EX_TCK					
	EX_TDO					
	1-46	DVP_SDA	46	IOT13B	LPLL_C_IN	4.7K - 2.5V
	1-44	DVP_SCL	44	IOT17B	GCLKC_0	4.7K - 2.5V
	1-43	DVP_VSYNC	43	IOT17A	GCLKT_0	
	1-42	DVP_HSYNC	42	IOT20B	GCLKC_1	
	1-41	DVP_PCLK	41	IOT20A	GCLKT_1	
	1-40	DVP_D9	40	IOT26B		
	1-39	DVP_D8	39	IOT26A		
	0-01	FLASH_CLK	1	IOT10A		
	0-09	FLASH_HOLD	9	IOT5B	DONE	4.7K - 3.3V
	1-47	FLASH_MISO	47	IOT11B		
	0-02	FLASH_CS	2	IOT10B		4.7K - 3.3V
	1-48	FLASH_MOSI	48	IOT11A		
	0-08	FLASH_WP	8	IOT4B		4.7K - 3.3V
	2-35	HDMI_TX2_P	35	IOR2A		
	2-34	HDMI_TX2_N	34	IOR2B		
	2-32	HDMI_TX1_P	32	IOR1A		
	2-31	HDMI_TX1_N	31	IOR1B		
	GND					

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Category	Kiwi Nano4K	Signal Name	Pin No.	Pin Name	Alternate Function	Pull-up
	GND					
	1V8					
	2V5					
	3V3					
	GND					
	TDI/0-03	JT DI	3	IOT2A		
	TMS/0-06	JTMS	6	IOT3A		
	TCK/0-07	JTCK	7	IOT3B		
	TDO/0-04	JTDO	4	IOT2B		
	2-33	DVP_XCLK	33	IOR9B	GCLKT_2	
	0-09	LED1	10	IOB7A	MODE	
	3-13	LED2	13	IOB4A		
	3-14	KEY1	14	IOB4B		100K – 1.8V
	3-15	KEY2	15	IOB5A		100K – 1.8V
	3-16	DVP_D6	16	IOB6A		
	3-17	DVP_D2	17	IOB6B		
	3-18	DVP_D5	18	IOB13A		
	3-19	DVP_D3	19	IOB13B		
	3-20	DVP_D4	20	IOB16A	GCLKT_5	
	3-21	DVP_D1	21	IOB16B	GCLKC_5	
	3-22	DVP_D0	22	IOB22A	GCLKT_4	
	3-23	DVP_D7	23	IOB22B	GCLKC_4	
	2-27	HDMI_TXC_N	27	IOR17B		
	2-28	HDMI_TXC_P	28	IOR17A		
	2-29	HDMI_TX0_N	29	IOR15B		
	2-30	HDMI_TX0_P	30	IOR15A		
	GND					

4 Consideration

Considerations for the use of the development board:

1. Handle with care and pay attention to electrostatic protection.
2. When downloading bitstream files to internal flash or external flash, set the MODE pin state to the correct configuration value, please refer to: <https://cdn.gowinsemi.com.cn/UG290E.pdf>
3. Power input 5V